

Substitute or Complement? Generative AI Use and Student Achievement: Evidence from PISA 2025

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Abstract

Generative AI can substitute for a student's effort, writing the essay, or complement it, explaining the material so they can write it. In PISA 2025, 54.2 percent of 15-year-olds in OECD countries use a chatbot at least weekly for school tasks. Comparing schoolmates with similar backgrounds, we find that more AI use is associated with lower achievement. Holding total use fixed, drafting and summarizing with AI are associated with lower scores, while research and learning go with higher ones. The associations are cross-sectional, but selection on ability alone would predict one direction for every purpose of AI use.

Keywords: artificial intelligence, skill formation, student achievement, PISA.

JEL codes: I21, J24, O33, C83.

1 Introduction

Nowadays, a 15-year-old with an essay assignment and a chatbot faces a choice that no earlier classroom technology offered. They can ask for the output, having the chatbot draft the essay, or for an input, having it explain the material before writing the essay themselves. The first substitutes for the effort that produces learning; the second complements it. The two acts look identical to a survey question about how often the student uses AI for school work, and they pull in opposite directions on what the student learns.

PISA 2025 is the first cycle of the assessment to ask 15-year-olds how they use AI, and it arrives at a charged moment. The same cycle records some of the steepest falls in OECD achievement in the assessment's history, with reading falling furthest (OECD, 2026a), and public discussion has turned to whether chatbots in schools are part of the explanation (The 74, 2026; Dao, 2026). What students do with AI is therefore central to reading the decline. The OECD's first report on these data shows the raw pattern, question by question and without controls: students who use chatbots score below those who do not (OECD, 2026a). Every purpose is correlated with total use, so each can look negative in the raw data even if they pull in opposite directions once total use is held fixed. This paper holds it fixed. We separate how much students use AI, which we call the level of use, from what they use it for, which we call its purpose, and estimate both within schools and net of socioeconomic background.

We find four things. The first concerns the level of use. A one standard deviation increase in our chatbot index is associated with 3.8 fewer PISA points in mathematics, 5.7 in reading and 6.1 in science within countries, and with 5.8, 7.6 and 7.9 fewer points when students are compared with their own schoolmates. Controlling for socioeconomic status makes these gradients steeper rather than flatter, moving the mathematics coefficient from -1.26 to -3.67 . That is what a normal good does to a cross-sectional gradient, and it means that analyses without background controls understate the association by nearly a factor of three.

The second finding, and the central one, is that this average hides a sign reversal. When the four questions on chatbot use, how often the student uses one to summarize a text, to research a topic, to draft writing and to help them learn, enter the regression together, so that each is read holding the other three fixed, using

AI to draft writing or to summarize a text is associated with lower mathematics scores, of -7.58 and -5.93 points per standard deviation of each question, about 1.4 response categories, while using it for preliminary research or to help one learn is associated with higher scores, of $+3.21$ and $+6.32$. The dividing line is exactly the line between AI producing the work and AI helping the student who does it, and the split survives when we compare students within the same school. A decomposition into two uncorrelated components tells the same story: the first, which captures the level of use, carries -3.67 points per standard deviation in mathematics, and the second, which contrasts learning with producing, carries $+7.15$. The components are uncorrelated by construction, so each margin stands on its own: the level gradient is net of the mix of purposes, and the purpose coefficient is net of how much AI is used.

Third, the negative side is strongest where a language model can best stand in for the student. Across specifications, the negative coefficients are 1.3 to 1.6 times larger in reading and science than in mathematics, which is the ordering substitution predicts, whereas a simple story of lower-achieving students reaching for chatbots would produce the same coefficient in every subject.

Finally, we ask where the positive margin is larger. We estimate the level and purpose slopes separately for each of 35 school systems, comparing students with their schoolmates, and then relate those slopes to features of each system. The purpose slope is larger in systems where students report more teacher support, by $+5.33$ (1.47) points per unit of a standardized teacher-support index whose country means span about one unit, and this holds after conditioning on how common AI use is and on average achievement. The level slope is steeper where more of the use happens at school, by -6.02 (2.38) points per response category of the gap between use at school and outside school, in the 22 countries where that split is measured. Both are patterns across systems: within a country, the same interactions at the school level are close to zero.

Reaching these estimates first requires solving a measurement problem that every study using these questions will face. Our index of the level of use, the first principal component of the four chatbot questions, is nearly their sum, and a sum gives the same value to a student who uses AI moderately for every purpose and to one who ticks the same box four times without reading. Such straight-lining is common, at 34.8 percent of students, and straight-liners score 18 to 59 points below students who reach the same total through different answers; the index the OECD distributes with the assessment is exposed in the same way. Every specification therefore controls for response style, and figures group students by the index only after that adjustment.

These are associations from a single cross-section of self-reported use, and the reverse story is obvious: a student who cannot write the essay has every reason to ask a chatbot to draft it. That story, however, makes a prediction the data reject. If struggling students turn to AI because they are struggling, they should turn to it for research and for learning as well, and every purpose should then be associated with lower scores. The reversal we find is the opposite of what that kind of selection produces, which is as far as a cross-section can go. Section 8 discusses each threat in turn.

The paper speaks to three literatures. Research on technology in schools has repeatedly found that access to computers and time spent on them are weakly or even negatively related to achievement (Angrist and Lavy, 2002; Machin et al., 2007; Malamud and Pop-Eleches, 2011; Fairlie and Robinson, 2013), while well-designed, targeted uses can produce large gains (Muralidharan et al., 2019). The common thread is that what matters is how technology is used rather than whether it is available (Falck et al., 2018; Escueta et al., 2020; Bulman and Fairlie, 2016; Fuchs and Woessmann, 2004; OECD, 2015), and we carry that distinction into the era of generative AI, where the tool can now produce the graded work itself. Experimental studies of generative AI already contain both of our signs. An unrestricted chatbot raises performance while students have it but harms learning once it is taken away (Bastani et al., 2025), whereas a chatbot designed as a tutor, with scaffolds that withhold the answer, outperforms active-learning instruction (Kestin et al., 2025). Evidence from the workplace documents the substitution side (Noy and Zhang, 2023; Brynjolfsson et al., 2025), and research on cognitive offloading explains why it can crowd out learning (Risko and Gilbert, 2016). Those studies identify effects for particular tasks and small samples; we provide the sign and

order of magnitude of the same contrast across 35 school systems. Finally, work on response styles in international assessments shows that self-reported scales are not comparable across students or countries without adjustment (Krosnick, 1991; Buckley, 2009; Meade and Craig, 2012; Vonkova et al., 2018).

2 Substitutes, complements, and skill formation

To fix ideas, think of achievement as the output of a cumulative production function in the student’s past inputs and own effort (Todd and Wolpin, 2003; Cunha and Heckman, 2007), and consider a single school task. The student supplies effort e , which produces two things: the graded output q that the teacher grades, and an increment L to skill, which is what the assessment measures. AI use a can enter this process in two ways, and the four PISA items happen to cover both.

In the first, AI enters the production of the output, $q = f(e + \phi a)$ with $\phi > 0$, so the chatbot supplies part of what the student would otherwise have supplied. Drafting a writing assignment and summarizing a text are of this kind, since what the model produces goes straight into the graded product. A student who can reach a given q with less effort will put in less, and because L depends on e rather than on q , learning falls. This is cognitive offloading in the sense of Risko and Gilbert (2016) and time displacement in the sense of Malamud and Pop-Eleches (2011), and it is the channel that Bastani et al. (2025) isolate when they take the tool away.

In the second, AI enters the production of skill directly and complements effort, $L = g(e, a)$ with $\partial^2 g / \partial e \partial a > 0$: the chatbot explains a step, answers a question or points to a source. Preliminary research and “help me learn” belong here. Instead of replacing effort, AI makes each unit of it more productive, which is the mechanism behind the gains in Muralidharan et al. (2019) and Kestin et al. (2025).

This distinction yields five predictions, and the rest of the paper is organized around testing them. The first and most direct is a sign reversal: holding total use fixed, purposes that point AI toward producing the output should be associated with lower achievement, and purposes that point it toward supporting the student with higher achievement. This is a statement about the mix of uses rather than their level, and it can be tested because the two can be separated.

The second concerns subjects. Substitution should be strongest where the model’s output is closest to the graded product, and current models stand in far better for written work than for mathematical reasoning, so the negative coefficients should be larger in reading and science than in mathematics. The framework’s subject ordering is specific to the substitution side.

The other three predictions are less about the mechanism itself. The third follows from AI use being a normal good: higher-status students both use it more and score higher, so controlling for socioeconomic status should make the gradient steeper rather than flatter. The fourth concerns what a complementarity needs: if steering AI toward learning takes teacher time and instruction, learning-directed use should be more strongly associated with achievement where students report more teacher support. The fifth describes the obvious alternative. If the association arose only because lower-achieving students reach for chatbots, they would reach for them for every purpose and all four items would be negative, so a sign reversal is where the cross-section can tell the two accounts apart.

3 Data and measurement

PISA assesses 15-year-olds in mathematics, reading and science on a scale with an OECD standard deviation of roughly 100 points, and the 2025 student file covers 755,721 students in 90 education systems (OECD, 2026b). Each student sees only part of the test, so proficiency is represented by ten plausible values drawn from each student’s posterior distribution (Mislevy, 1991; von Davier et al., 2009).

The 2025 student questionnaire contains eight items on artificial intelligence, in three groups that measure different things: four chatbot items, two technology items, and two others. The four chatbot items form a battery, fielded in every country, that asks how often the student uses AI chatbots for school work to summarize a text, to conduct preliminary research, to draft writing assignments and to help them learn, on a

five-point scale from never to daily. The two technology items, in the optional questionnaire on information and communication technology, ask how often the student used AI tools at school and outside school, so they exist only for some countries. The remaining two ask how often lessons involved assessing the quality of AI-generated information, and whether the student is interested in learning about machine learning and AI. Online Appendix D gives the exact wording and the weighted distribution of answers.

Use is already widespread, and it leans toward tasks where AI can replace student effort as easily as support it. Between 29.0 and 45.6 percent of students use a chatbot at least weekly for each of the four tasks, most often to help them learn and least often to draft writing. Across the four, 54.2 percent report at least weekly use for at least one task, and only 13.7 percent say they never use a chatbot for any of them.

The four chatbot items correlate between 0.60 and 0.69 with one another; the two technology items correlate 0.61 with each other and 0.40 to 0.46 with the chatbot block; and the remaining two stand apart from both. Assessing AI output in lessons correlates only 0.31 to 0.35 with the chatbot items and more strongly with AI use at school, as one would expect of an activity the teacher assigns rather than one the student chooses, and interest in machine learning is almost unrelated to everything else, with its largest correlation at 0.21. We keep both as separate variables outside the index.

We build two indices from weighted principal components of the correlation matrix, standardizing the items with senate weights and scoring component k as $f_{ik} = z'_i v_k / \sqrt{\lambda_k}$, so that each index has weighted mean zero and standard deviation one, with higher values meaning more AI use. The *chatbot* index uses the four universal items and covers all 38 countries; the *broad* index adds the two technology items and covers the 24 countries that fielded them. On the chatbot index, raising all four answers by one category, from monthly to weekly use for instance, adds 0.84 standard deviations. The first component of the chatbot items captures 74.0 percent of their variance, Cronbach's α is 0.882 (Cronbach, 1951), the loadings run from 0.829 to 0.876, and the conventional eigenvalue rule retains only that component (Kaiser, 1960). The index correlates 0.979 with the one distributed with the assessment and 0.960 with the broad index. The second component is small, below that rule, but it is exactly the contrast of Section 2: it loads +0.527 on “help me learn” against -0.275 on summarizing and -0.262 on drafting. Online Appendix E gives the construction, the loadings and a country-by-country check.

The near-equal loadings have a less welcome consequence. They make the first component close to the simple sum of the four items, and a simple sum is exactly what careless responding distorts. Respondents who satisfice tend either to pick the same option again and again or to answer with little attention to content (Krosnick, 1991; Meade and Craig, 2012), and both habits are common here. Among students who answer all four chatbot items, 34.8 percent give the same answer to every one. When we compare these straight-liners with students who reach the same total through different answers, measuring both relative to their country's mean, the gaps are large and systematic. At a total of 8, straight-liners score 37.4 points below their country mean while mixed responders score 21.9 points above it, a difference of 59.3 points; the gap is 31.4 points at a total of 12 and still 18.3 at a total of 16.

Each item is scored from 1 to 5, so a student who gives the same answer to all four has a total of 4, 8, 12, 16 or 20, and straight-liners sit only at those five totals. Since they score far below other students with the same total, the average score at 8, 12 and 16 dips below the averages at the neighboring totals, and any statistic that groups students by the item sum, a binned scatter for instance, inherits these dips, which reflect response style rather than AI use. Two rules follow, and we use both below. Every specification controls for response style, using a straight-lining indicator together with the within-student standard deviation across the standardized items and its square; and we bin only the adjusted index. Figure 1 shows the problem directly, and shows that the problem survives the obvious fix of dropping straight-liners. The deeper reason is that the index is built from four five-point answers, so each group of students mixes different answer patterns, and those patterns carry information about achievement of their own (Online Appendix F).

Both response-style terms are functions of the four items themselves, so we also measure inattention independently of the answers. The assessment's timing file records the seconds each student spent on the

screen holding the four items; the median is 19 seconds, and the fastest tenth of students within each country, under about eight seconds, are too quick to have read four stems. Adding this indicator to every specification, or dropping the students it flags, moves the within-school level coefficient in mathematics from -5.77 to -6.26 and -6.36 , and the purpose coefficient of Section 6 from $+4.44$ to $+3.96$ and $+4.01$. Across the other definitions of Online Appendix B, built from self-contradictory answers, straight-lining elsewhere in the questionnaire and self-reported effort, the level coefficient stays between -4.9 and -7.8 and the purpose coefficient between $+3.3$ and $+5.3$.

Figure 1: Average score by group of the AI-use index



Notes: Students are sorted by the unadjusted AI-use index (the first principal component of the four chatbot questions, in standard deviations) and split into groups of about 5 percent; each point is a group's average index and score. Orange: all students; students with identical answers stay together, so this series has 18 groups, the first being the 14 percent who answer “never” to all four questions. Blue, with its 95 percent interval: excluding the 34.8 percent who give the same answer to all four (20 groups). Every group holds at least 4,800 students, so the jumps are systematic: they reflect which answer patterns fall into each group. Senate-weighted, OECD countries; Figure 3 shows the adjusted version.

Our estimation sample covers the 38 OECD members in PISA 2025, identified by the file's own OECD indicator, for a total of 315,314 students. All 38 fielded the chatbot battery and 24 fielded the optional technology questionnaire. Three countries drop out of the regressions: the socioeconomic status index is missing for Costa Rica, and school identifiers for Iceland and Luxembourg, since both assess close to their entire eligible population. That leaves 249,077 students in 10,403 schools and 35 countries, and we use this same sample in every specification so that the rows of Figure 2 can be compared directly. Online Appendix G lists the countries.

4 Empirical strategy

Our strategy exploits two features of the data. First, PISA samples whole schools, so students can be compared with schoolmates who share teachers, curriculum and school policy on AI. Second, the chatbot questions ask about purposes, so the four can be entered together and each read at a fixed total, the contrast Section 2 calls for. The estimates are associations, and Section 8 discusses the selection within schools that remains. Our baseline specification is

$$y_{ic} = \beta AI_{ic} + \gamma' X_{ic} + \alpha_c + \varepsilon_{ic},$$

where y_{ic} is achievement, AI_{ic} is the standardized index and α_c is a country or school fixed effect. We add controls one step at a time: first the index alone, then country fixed effects so that the comparison is within country, then the assessment's index of economic, social and cultural status, then the response-style terms of Section 3, and finally school fixed effects, absorbed by demeaning within school, so that each student is compared with their schoolmates. The index is standardized, so β is measured in PISA points per standard

deviation of reported AI use. In Section 6 we replace the single index with either the four items entered together or the first two principal components, which splits β into the level and purpose margins of Section 2.

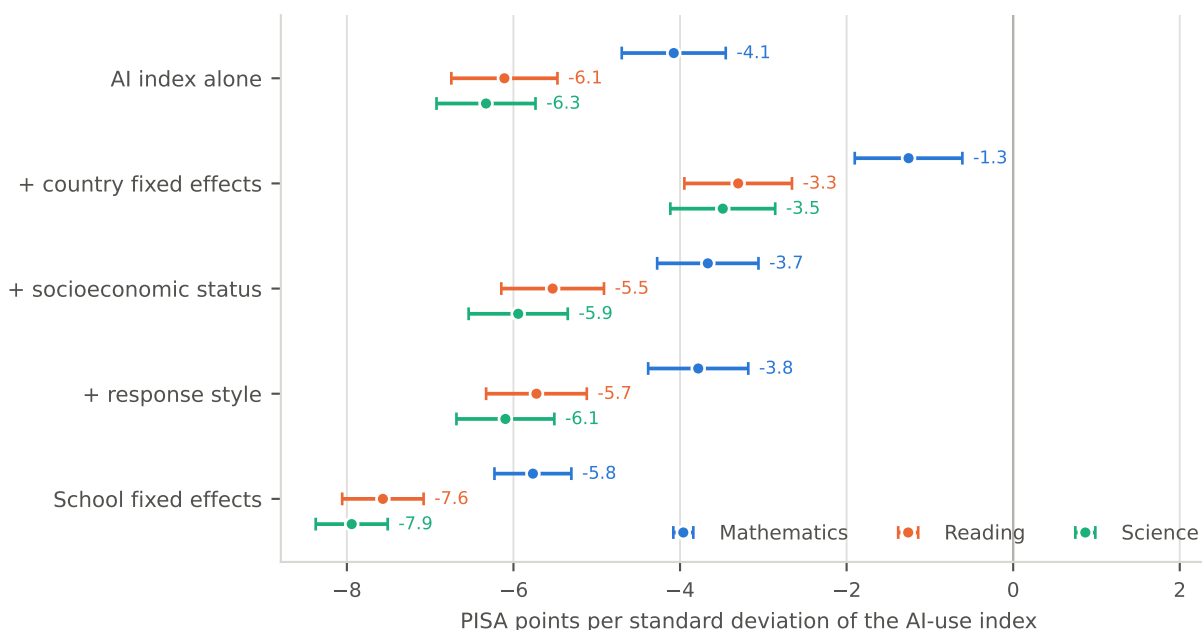
All estimates use senate weights, so that each country counts equally, renormalized within country on each estimation sample. Each achievement measure is the average of the domain's ten plausible values, and standard errors combine the 80 Fay replicate weights with Rubin's imputation variance across plausible values (Judkins, 1990; Rubin, 1987; OECD, 2024). The renormalization matters because the share of students answering the items ranges from 53 percent in the United States to 97 percent in Korea, which would otherwise give Korea 85 percent more weight. Under school fixed effects we also redo the within-school demeaning for every replicate weight, since a weighted demeaning depends on the weights.

To show the data behind each coefficient we use residualized binned scatters: both the index and achievement are regressed on the specification's controls, the residuals are grouped into twenty equally weighted bins, and the line through the bins has the regression coefficient as its slope. Residualizing also removes the binning problem of Section 3, because the residual of a lumpy variable on continuous controls is itself continuous, and the bin means become smooth. The usual caution about reading binned scatters as informal inference applies (Cattaneo et al., 2024).

5 The level margin

This section estimates the level margin, the difference in achievement, in PISA points, between students one standard deviation apart on the chatbot index, a little more than one response category on every question, and shows how it moves as the comparison narrows from all students to schoolmates. Figure 2 reports the coefficient under each specification, and Online Appendix Table A.1 the full estimates.

Figure 2: AI use and achievement across specifications



Notes: Coefficient on the standardized chatbot index, in PISA points per standard deviation, with 95 percent intervals that combine 80 Fay replicate weights with Rubin imputation variance across ten plausible values. Senate-weighted with within-sample renormalization. The first four rows add controls cumulatively; the last replaces country with school fixed effects. Every row uses the same 249,077 students in 35 countries. Response-style controls are a straight-lining indicator and the within-student standard deviation across the standardized items and its square. Exact estimates and standard errors are in Online Appendix Table A.1.

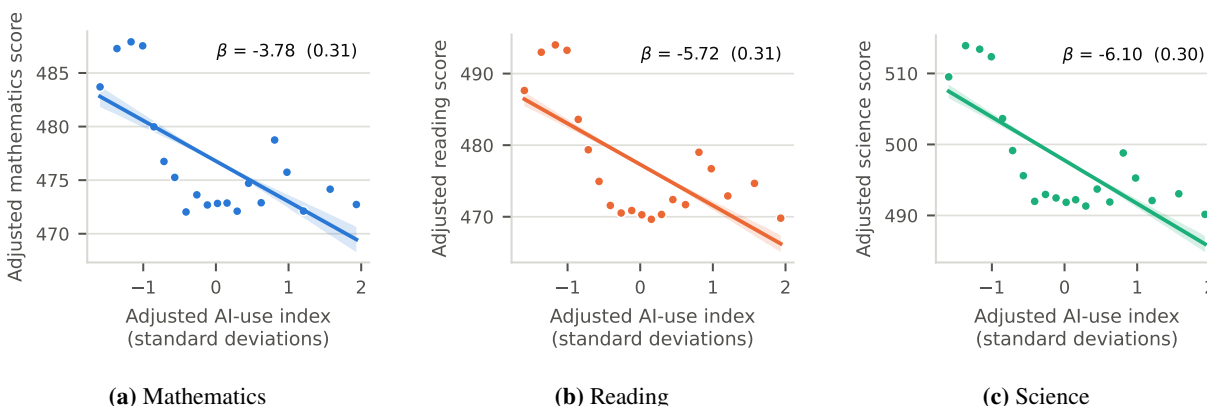
With the index alone, the association is negative in every domain: -4.08 points per standard deviation in

mathematics, -6.11 in reading and -6.33 in science. Adding country fixed effects shrinks the mathematics coefficient to -1.26 , so much of the raw association reflects differences between countries, and what remains compares students within the same country.

The third row tests the third prediction, and the data move in the predicted direction. Adding socioeconomic status makes the coefficients *more* negative, from -1.26 to -3.67 in mathematics and from -3.30 to -5.53 in reading. This is suppression rather than the usual kind of confounding, and it is what a normal good produces: higher-status students report more AI use and also score higher, so leaving status out masks the gradient. The response-style controls move the coefficients a little further, and so does the demographic block of Online Appendix C, which takes the within-country mathematics coefficient from -3.78 to -4.40 : like socioeconomic status, background works as a suppressor here.

The last row replaces country with school fixed effects, which removes school quality, curriculum, the local mix of students and any school-level policy on AI. The association becomes substantially *stronger*, at -5.77 , -7.57 and -7.94 , and its standard errors are the smallest in the figure because comparisons within schools are more precise. Adding pre-determined demographics leaves these estimates within a third of a point, and adding engagement and effort scales, which sit inside the mechanism of Section 2, lowers the mathematics coefficient to -4.72 with every sign intact (Online Appendix C). Sorting across schools was therefore hiding part of the gradient rather than creating it, the opposite of what a pure school-selection story would predict.

Figure 3: Average score by group of the AI-use index, after adjustment



Notes: Both the AI-use index and the score are adjusted for country fixed effects, socioeconomic status and the response-style terms, that is, replaced by their difference from what those variables predict (plus the mean). Students are sorted by the adjusted index into 20 groups of about 5 percent; each point is a group's average. The line's slope is the coefficient in the fourth row of Figure 2, with its 95 percent interval shaded. Higher index values mean more AI use.

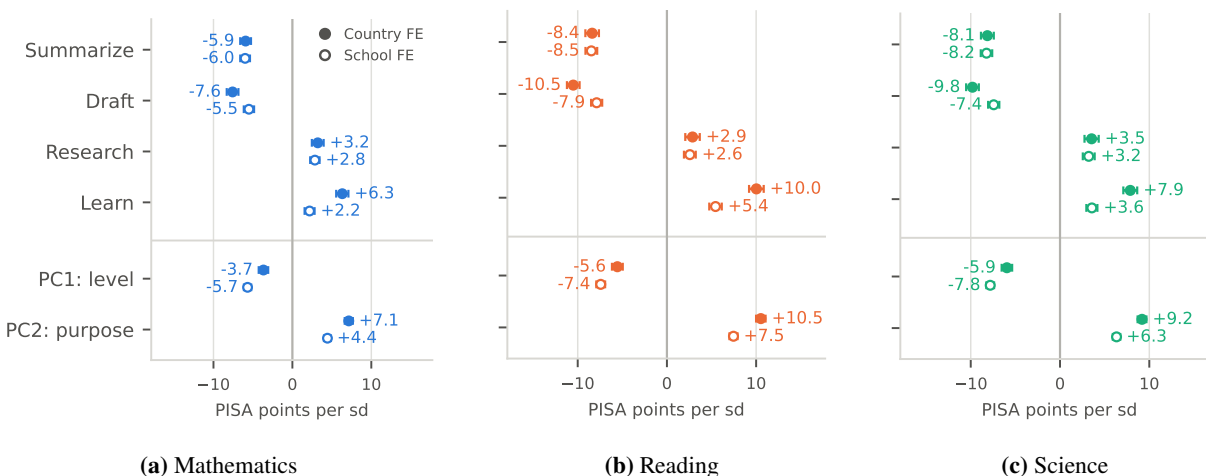
Figure 3 shows the data behind the full-controls coefficient. Grouped on the unadjusted index, average scores were jagged (Figure 1); once both variables are adjusted for the controls, the scatter is tight and close to monotone. The figure also shows that the relationship is convex, steep across the lower part of the index and flatter above its mean. A quadratic term under the full controls confirms this, at $+6.10$ in mathematics, $+9.39$ in reading and $+8.25$ in science, so a single linear coefficient understates the gradient at the low end and overstates it at the high end.

The level results are also consistent with the second prediction. Depending on the specification, the reading and science coefficients are 1.3 to 1.6 times the mathematics coefficient. That is the ordering substitution implies, since a language model stands in far better for written work than for mathematical reasoning, and a pure ability-sorting story would produce the same coefficient in every subject.

6 Substitution and complementarity

So far every estimate has been about how much students use AI. The first prediction is about what they use it for, and the items allow two ways of asking. Entering the four chatbot items together reads each one holding the other three fixed, which is the empirical counterpart of holding total use constant. Entering the first two principal components instead splits the association into a level dimension and a purpose dimension that are uncorrelated by construction. Each item is standardized, so its coefficient is in PISA points per standard deviation of that item, about 1.4 response categories. The purpose component is standardized too: shifting one category of use from drafting or summarizing to help with learning, at the same level of use, raises it by about 1.3 standard deviations and leaves the level component essentially unchanged.

Figure 4: The association by kind of AI use



Notes: PISA points per standard deviation of the item or component (one standard deviation of an item is about 1.4 response categories), with 95 percent intervals, for 249,077 students in 35 OECD countries. The top four rows enter the four chatbot items together, so each is read holding the other three fixed (variance inflation factors 2.0 to 2.5); the bottom two enter the first two principal components together, which are uncorrelated by construction. Filled markers condition on country fixed effects and hollow markers on school fixed effects; both include socioeconomic status and the response-style terms. Exact estimates and standard errors are in Online Appendix Table A.2.

Both approaches show the reversal (Figure 4). Under country fixed effects, drafting writing and summarizing a text are associated with -7.58 and -5.93 points in mathematics and -10.50 and -8.38 in reading, while preliminary research and help with learning are associated with $+3.21$ and $+6.32$ in mathematics and $+2.87$ and $+10.03$ in reading. Every one of these coefficients is significant well beyond conventional levels, and the split falls exactly where the framework puts it: the two items whose output feeds into the graded product are negative, and the two that support the student producing it are positive. The component version agrees, with the level component at -3.67 and the purpose component at $+7.15$ in mathematics.

Three aspects of these estimates deserve comment. The first is that the level result stands on its own, whatever the mix of purposes. The two components are uncorrelated, so adding the purpose dimension leaves the level coefficient almost unchanged, at -3.67 against -3.78 in Figure 2. Holding the mix of uses fixed, more use is still associated with lower achievement. The split holds with the demographic block of Online Appendix C, and narrows while keeping its sign when the engagement and effort scales, which are part of the substitution channel, are added: within schools drafting moves from -5.51 to -3.54 and learning from $+2.18$ to $+1.35$.

The second is that the substitution side follows the domain ordering consistently, which is what the framework expects. Within schools, drafting is associated with -5.51 points in mathematics and -7.88 in reading, and summarizing with -5.99 and -8.47 , ratios of about 1.4. The positive items are less uniform:

preliminary research is of similar size in every subject, and help with learning is larger in reading than in mathematics. In the framework, substitution operates through the graded product, whose linguistic content differs sharply across subjects, while the complementary channel operates through comprehension, which is common to all three.

The third, and the most informative, is how the two sides respond to school fixed effects. The sign split survives within schools: drafting and summarizing still carry -5.51 and -5.99 in mathematics, and research and learning $+2.85$ and $+2.18$. However, the positive coefficients shrink sharply, the learning item most of all, falling from $+6.32$ to $+2.18$ in mathematics, while the negative ones barely move. On the framework's reading, substitution is a choice the student makes and so travels with the student, whereas complementarity depends on the setting, so absorbing the school absorbs much of it; Section 7 tests that reading directly. Both margins carry independent associations of comparable size: with country fixed effects the purpose coefficient is the larger in every domain, and with school fixed effects the level coefficient is the larger in mathematics and science.

7 Where complementarity is realized

If the positive margin depends on the setting as well as the tool, it should be larger where the setting provides more of what it takes to steer AI toward learning. That is the fourth prediction: the purpose slope should be larger where teacher support is higher. We look for it at two levels, across countries and across schools within a country.

Across countries, we proceed in two steps. First, for each country c we estimate the specification of Section 6 within school,

$$y_{isc} = \beta_c^L \text{PC1}_{isc} + \beta_c^P \text{PC2}_{isc} + \gamma_c' X_{isc} + \alpha_{sc} + \varepsilon_{isc},$$

where i is a student, s a school and α_{sc} a school fixed effect. This gives each country a level slope $\hat{\beta}_c^L$ and a purpose slope $\hat{\beta}_c^P$. Second, we regress those slopes on a characteristic Z_c of the school system,

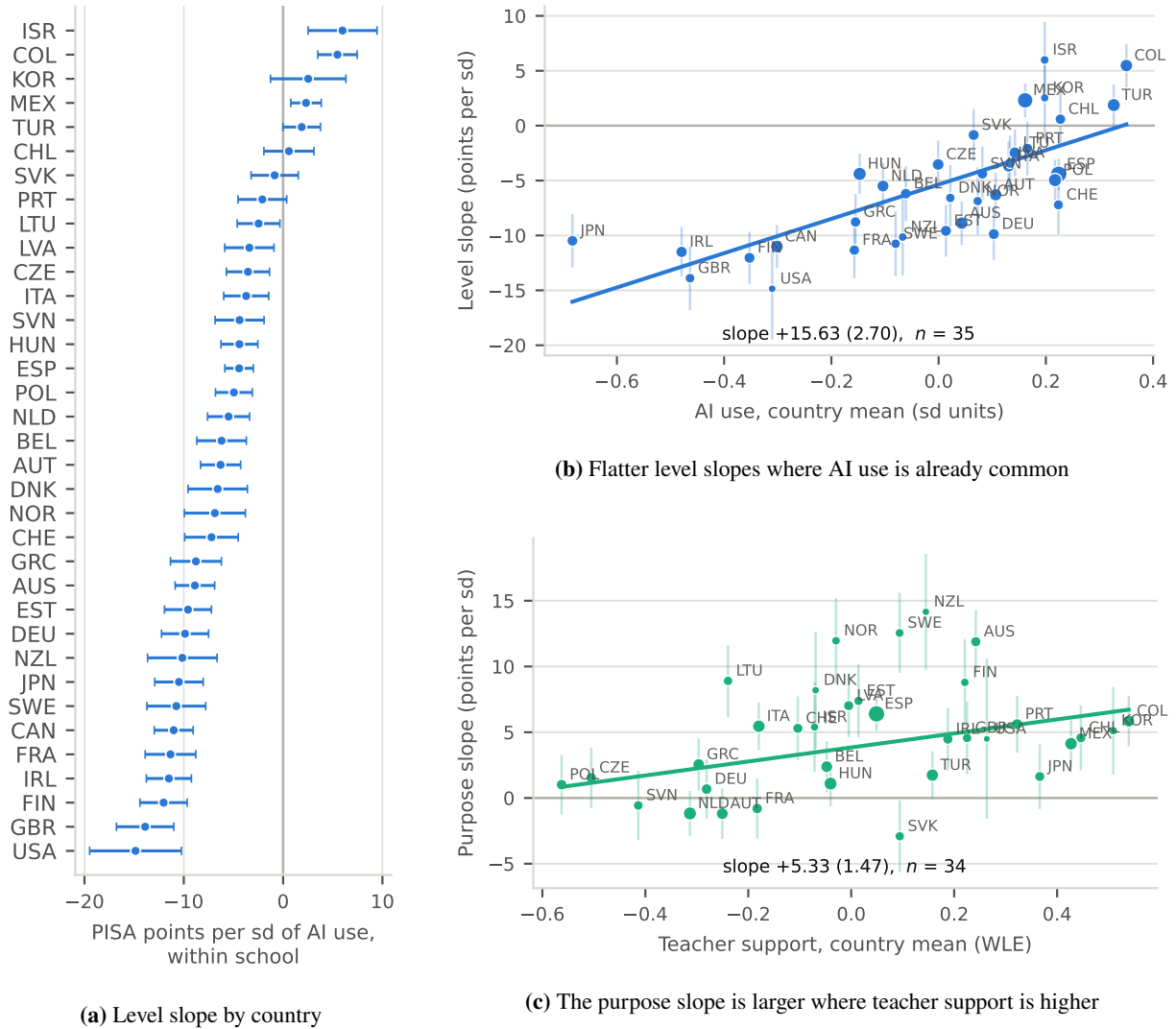
$$\hat{\beta}_c^P = a + b Z_c + u_c,$$

and likewise for $\hat{\beta}_c^L$, weighting each country by the inverse of the squared standard error of its slope, so that precisely estimated slopes count more.

The level slope, the achievement gap between schoolmates one standard deviation apart on the chatbot index, varies a great deal across countries, and very little of that variation is sampling noise: sampling error accounts for about 5 percent of it in mathematics and 3 percent in reading (Figure 5, panel a). In mathematics it runs from -14.9 points in the United States and -13.9 in Britain to $+6.0$ in Israel, and it is significantly positive in Israel, Colombia and Mexico; in reading it spans -22.8 to $+12.4$. The purpose slope, the achievement difference that goes with a one-standard-deviation shift of use toward help with learning, is positive in 30 of the 35 countries and runs from -2.9 to $+14.2$ points in mathematics.

Three characteristics of school systems line up with these slopes. First, the purpose slope is larger where students report more teacher support (panel c). The teacher-support index is standardized across OECD students, and its country means run from -0.56 in Poland to $+0.54$ in Colombia; across the 34 countries with the measure, a system one unit higher has a purpose slope larger by 5.33 (1.47) points. Holding fixed how common AI use is, the estimate is $+5.23$ (1.47), and holding fixed mean achievement as well, $+5.91$ (1.72) (Online Appendix Table A.3, Panel B). That is the fourth prediction. Second, level slopes are steeper where more of the use happens at school. The gap between how often students use AI tools at school and outside school, in response categories, runs from -0.69 in Turkey to $+0.17$ in the Netherlands across the 22 countries that fielded the technology questionnaire, and a gap one category larger goes with a level slope 6.02 (2.38) points more negative. Third, level slopes are flatter where AI use is common (panel b). Country means of the chatbot index run from -0.68 in Japan to $+0.35$ in Colombia, and a system whose mean is one standard

Figure 5: System heterogeneity in the AI gradient



Notes: PISA 2025, 35 OECD countries with at least 1,500 usable students. Panel (a) gives each country's within-school level slope with its 95 percent interval, sorted. In panels (b) and (c) marker area is proportional to the precision of the country's first-step estimate and the fitted line is weighted by that precision.

deviation of student use higher has a level slope 15.63 (2.70) points flatter. Part of this is mechanical, since where few students use AI the users are a selected minority, but re-estimating every country's slope among users only still leaves +11.90 (2.33) (Panel B of the same table); we report this pattern as a description of the data. Panel A of that table re-estimates all three relationships with random-effects weights, classical standard errors, one country dropped at a time and rank correlations, and each keeps its sign and size.

Across schools within a country, we ask whether the same characteristics matter. We interact each component with the mean of the characteristic among the student's schoolmates, with school fixed effects as before (Online Appendix Table A.4). The flattening where AI use is common reappears, at +3.35 (0.92) points per unit of the school's mean index; school means vary within a country with a standard deviation of 0.31, so schools one standard deviation apart differ by about one point in the level slope. The other two interactions are close to zero: -0.86 (0.67) for teacher support measured at the school and +0.28 (0.21) for the student's own report, and +0.59 (0.70) for school-based use. Teacher support and the place of use are therefore features of whole systems, while the prevalence of AI use goes with a flatter level slope at both

levels.

These are conditional correlations, across at most 35 countries and 22 for the place of use. Teacher support is also reported by students, so it may carry some of the response style that the student-level specifications control for, and countries differ in how they use such scales (Buckley, 2009; Vonkova et al., 2018). We report the teacher-support result because it is the prediction the framework makes and, were it causal, it would point to an input that policy can change.

8 Interpretation and threats

A standard deviation of reported AI use goes with roughly 0.06 to 0.08 standard deviations of achievement. A student who never uses AI sits about one and a half standard deviations of the index below the mean, so on the school-fixed-effect estimate the gap between the two is on the order of eight to twelve PISA points. That is small next to a year of schooling, which the OECD puts at about 20 points, and it is sizeable next to the cross-country differences in achievement that carry most of the measured relationship between cognitive skills and growth (Hanushek and Woessmann, 2008): the standard deviation of mean mathematics across the 38 OECD countries is only 32 points.

Reverse causality is the most serious threat, and it bites hardest on the negative items: a student who cannot write the essay has the most direct reason to ask a chatbot to draft it, and drafting is the most negative of the four. Two facts limit how far it can go. A story in which low achievers turn to AI because they are struggling predicts negative coefficients for all four purposes, since a struggling student would also use the tool for research and for learning, and the reversal in Figure 4 is the opposite of what that story predicts. The same story also implies that absorbing school quality should weaken the association, because lower-achieving students are concentrated in lower-achieving schools, yet school fixed effects strengthen it. Both facts are descriptive, and selection that works purpose by purpose remains possible; only an experiment could settle it. What they show is that the pattern requires more than uniform sorting by ability.

Selection may also lie behind the convexity documented in Section 5. The convexity fits substitution, but it fits equally well a story in which the students who move first from never to occasional use are disproportionately those already struggling. The cross-section leaves the two observationally equivalent, and the convexity confines the linear coefficient to the middle of the index.

Measurement is a further concern. Use is self-reported. Careless answering is covered by the response-style controls and by the checks of Online Appendix B, and systematic misreporting remains: if lower-achieving students overstate their use, or stronger students understate drafting in particular, the item coefficients absorb it. The items also measure frequency. A student who reports daily use to “help me learn” may be getting explanations or answers, and the four labels are the only handle the questionnaire offers on that difference.

Finally, coverage varies by item. The technology items exist for 24 of the 38 countries, the socioeconomic index for all but Costa Rica, and school identifiers for all but Iceland and Luxembourg, so the school-fixed-effect estimates describe 35 systems, and the second stage 22 countries, where use can be split between school and home. Gender is recorded in the PISA sampling data as female or male. The estimates pool both groups, with gender among the controls of Online Appendix C, and differences by gender are a limitation of the analysis.

9 Conclusion

Using AI for school work is already the norm for 15-year-olds across the OECD, and the more useful question about it is what students use it for. Holding total use fixed and comparing schoolmates in PISA 2025, using AI to produce work is associated with lower achievement and using it to learn with higher achievement. The two margins are of comparable size, the negative one is largest in the subjects where a language model can best stand in for the student, and the positive one is larger in school systems where students report more teacher support.

Two implications follow. For policy, the question is less whether students should use AI than how. For research, indices built from these items, including the one released with the assessment, need explicit response-style controls and should be binned only after adjustment.

PISA 2025 also records some of the steepest recent falls in OECD achievement in the assessment's history (OECD, 2026a), and it is natural to wonder whether the two are linked. These estimates leave that question open: the decline began a decade before chatbots existed, every country was exposed at once, and the same period includes the recovery from the pandemic. Still, the margin we document is negative, it is largest in reading, and reading is where the recent fall is steepest, so whether AI use has contributed to the aggregate decline is a question these estimates make worth asking.

The causal effect of AI on learning is a question for the experimental and longitudinal work that must follow; what we have shown is that the pattern requires more than uniform sorting by ability. What these estimates offer that work is a benchmark across 35 school systems, built on a defensible index, with the relevant contrast isolated and the main measurement trap in the new items mapped.

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Data availability

The PISA 2025 data (OECD, 2026b) are publicly available from the OECD at <https://www.oecd.org/en/data/datasets/pisa-2025-database.html>. The code that reproduces all results will be deposited in a public repository upon publication.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to revise the text and check references. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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